

Purpose Financial: Loan Default Prediction

Prepared by: Daniel W. Livingston, Senior Data Scientist

1. Azure AutoML Configuration & Execution

The model training was orchestrated using the Azure Machine Learning SDK. The pipeline was configured to point directly to the data asset in Blob Storage, bypassing local memory constraints and ensuring scalable execution.

Key Configurations:

- **Optimization Metric:** 10-Fold Cross-Validation optimizing for `auc_weighted` to account for class imbalances.
- **Resource Guardrails:** A 60-minute timeout, a maximum of 15 trials, and a cap of 3 concurrent runs were enforced to strictly control cloud compute costs.
- **Featurization:** Auto-featurization was enabled to handle baseline imputations, scaling, and encoding across the pipeline.

Algorithm name ▾	Responsible AI ▾	AUC weighted ↓	Sampling ▾	Created on
VotingEnsemble		0.76784	100.00 %	Jul 14, 2026 10:52 AM
StackEnsemble		0.76723	100.00 %	Jul 14, 2026 10:52 AM
StandardScalerWrapper, XGBoostClassifier		0.76215	100.00 %	Jul 14, 2026 10:36 AM
MaxAbsScaler, LightGBM		0.75591	100.00 %	Jul 14, 2026 10:36 AM
SparseNormalizer, LightGBM		0.75334	100.00 %	Jul 14, 2026 10:36 AM
MaxAbsScaler, ExtremeRandomTrees		0.75130	100.00 %	Jul 14, 2026 10:36 AM
StandardScalerWrapper, LightGBM		0.74993	100.00 %	Jul 14, 2026 10:36 AM
SparseNormalizer, RandomForest		0.74777	100.00 %	Jul 14, 2026 10:36 AM
SparseNormalizer, XGBoostClassifier		0.74092	100.00 %	Jul 14, 2026 10:36 AM
SparseNormalizer, XGBoostClassifier		0.74061	100.00 %	Jul 14, 2026 10:36 AM
MaxAbsScaler, XGBoostClassifier		0.73555	100.00 %	Jul 14, 2026 10:36 AM
MaxAbsScaler, ExtremeRandomTrees		0.73376	100.00 %	Jul 14, 2026 10:36 AM
MaxAbsScaler, RandomForest		0.72827	100.00 %	Jul 14, 2026 10:36 AM
StandardScalerWrapper, XGBoostClassifier		0.72003	100.00 %	Jul 14, 2026 10:36 AM
MaxAbsScaler, RandomForest		0.71279	100.00 %	Jul 14, 2026 10:36 AM

Overview
Model
Metrics
Responsible AI (preview)
Data transformation (preview)
Test results (preview)
Outputs + logs
Images
Child jobs
Code
Monitoring

Refresh
Deploy
Download
Explain model
View generated code
Test model (preview)
Register model
Cancel
Delete
Compare

Properties

Status
Completed

Created on
Jul 13, 2026 3:37 PM

Start time
Jul 13, 2026 3:37 PM

Duration
1m 1.250s

Compute duration
1m 1.250s

Name
polite_sheep_t0gbpw4rx3_13

Script name
automl_driver.py

Created by
Dan Livingston

Job type
Command

Experiment
purpose_financial_loan_default

Environment
AzureML-ai-ml-automl:41

Arguments
None

See all properties
Raw JSON

See YAML job definition
Job YAML

Tags

mlflow.source.name : automl_driver.py

mlflow.source.type : JOB

model_explain_run_id : polite_sheep_t0gbpw4rx3_ModelExplain

model_explanation : True

Inputs

Input name: training_data

Data asset: azureml_polite_sheep_t0gbpw4rx3_input_data_TrainData:1

Asset URI: azureml:azureml_polite_sheep_t0gbpw4rx3_input_data_TrainData:1

Outputs

Output name: mlflow_log_model_1589189412

Model: azureml_polite_sheep_t0gbpw4rx3_13_output_mlflow_log_model_1589189412:1

Asset URI: azureml:azureml_polite_sheep_t0gbpw4rx3_13_output_mlflow_log_model_1589189412:1

Metrics

Accuracy
0.89784

AUC macro
0.76778

AUC micro
0.94690

AUC weighted
0.76778

Average precision score macro
0.61014

Average precision score micro
0.94396

View all other metrics

Description

Click edit icon to add a description

2. Champion Model Selection & Compliance Requirements

The winning pipeline from the automated trials was a **VotingEnsemble**, which consistently outperformed standalone XGBoost and LightGBM models. This ensemble was registered to the MLflow workspace registry as **Purpose_Champion**.

Fair Lending Compliance Note (ECOA): During the feature engineering phase, the **date_of_birth** and derived **age** features were intentionally scrubbed from the training dataset. This strict exclusion ensures the model complies with the Equal Credit Opportunity Act (ECOA) and broader fair lending regulations, preventing any algorithmic bias or disparate impact based on the applicant's age.

Default Directory > Purpose > Models > Purpose_Champion:1

Purpose_Champion:1 ☆

Details Versions Artifacts Endpoints Jobs Data Feature sets Responsible AI

Refresh Archive Use this model Download all Share model

Attributes

Name
Purpose_Champion

Version
1

Created on
Jul 14, 2026 12:38 PM

Created by
Dan Livingston

Type
MLFLOW

Created by job
[coral_bag_jfqjxls24m_13](#)

Asset ID
[azureml:/locations/eastus2/workspaces/a87cff33-fb98-4668-ba10-af76b2127761/models/Purpose_Champion/versions/1](#)

Tags

No tags

Properties

No properties

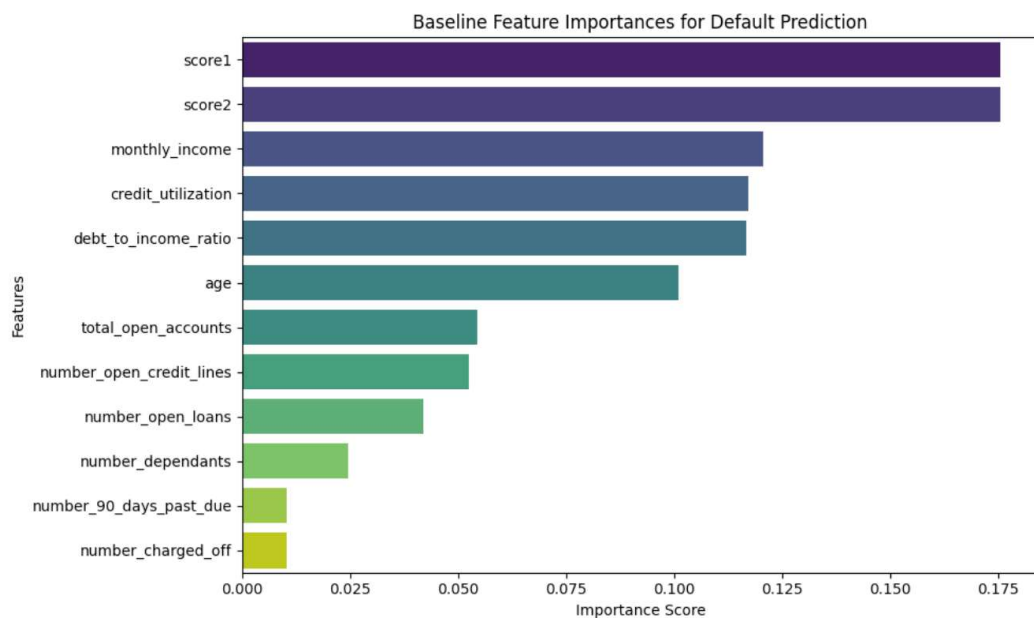
Description

Click edit icon to add a description

3. Feature Importances & Baseline Drivers

Prior to extracting deep interpretability metrics from the complex ensemble, a baseline Random Forest was trained on the engineered features. This baseline utilized a `class_weight='balanced'` parameter to actively counter the ~90/10 target class imbalance and establish preliminary, directional feature importances.

- **Primary Drivers:** Credit scores (`score1`, `score2`) and overall financial liquidity (`monthly_income`, `credit_utilization`) surfaced as the dominant predictive features.

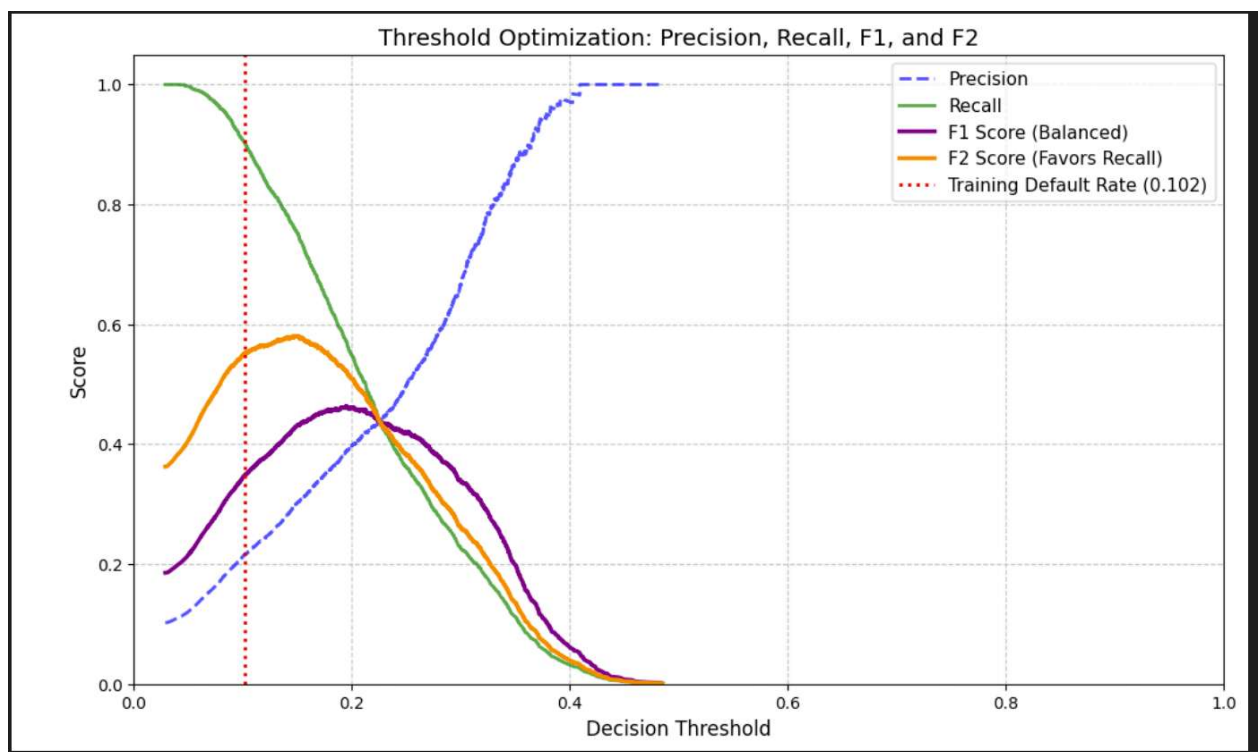


4. Decision Threshold Optimization

Evaluating Precision and Recall in isolation is insufficient for financial risk pipelines. To address the heavy class imbalance and align the model with business realities, probability distributions were mapped against Precision, Recall, F1, and F2 scores.

Operational Cutoff Selected: 0.15

- **Mathematical Justification:** This threshold aligns with the absolute peak of the F2 curve, prioritizing Recall over Precision.
- **Risk Mitigation:** At this sensitivity, the model successfully captures approximately 75% to 80% of actual defaults before they occur.
- **Business Context:** For early intervention and collections routing, the financial loss of a missed charge-off (false negative) is substantially more damaging than the operational friction of a proactive account review (false positive).



5. Model Interpretability & Risk Drivers (SHAP)

To bypass the "black-box" nature of the `VotingEnsemble` pipeline, SHAP (SHapley Additive exPlanations) was applied using a `KernelExplainer` wrapper. This provides necessary transparency into the macro-drivers of risk across the portfolio.

Reading the Summary Plot:

- Features are ranked top-to-bottom by overall predictive impact.
- Points pushed to the **right** indicate that the specific feature value increased the predicted probability of default.
- **Red points** indicate high values for that feature. For example, lower credit scores (blue) and higher credit utilization (red) strongly drive default risk upward.

